

Exercise: Designing a Data Management Strategy

Scenario:

You are a data architect for a retail company that operates online and offline stores. Your company collects data from various sources, including:

1. **Point-of-Sale Systems (POS)** (transactional data).
2. **Customer Feedback** from surveys and social media.
3. **Web Analytics** from the e-commerce platform.
4. **Supply Chain Data** from logistics and inventory systems.

Your task is to design a data management strategy by answering the following:

1. **Data Storage**
 - Which type of storage (relational database, data lake, or data warehouse) is suitable for each data source? Justify your answer.
2. **Data Integration**
 - Describe how you would integrate data from these sources using ETL/ELT processes. Include a brief flow diagram.

Suggested answers for the Exercise #4

1. Data Storage Strategy

Data Source	Recommended Storage	Comments
POS (Transactional Data)	Relational Database (RDBMS)	Ensures structured, ACID-compliant storage for transactions.
Customer Feedback (Surveys, Social Media)	Data Lake	Stores raw, unstructured data for sentiment analysis.
Web Analytics (E-commerce)	Data Warehouse	Structured for reporting and trend analysis.
Supply Chain (Logistics & Inventory)	Hybrid (Data Lake + Warehouse)	Raw logs (e.g. Geodata, TimeTrack, Condition) in a lake, structured summaries in a warehouse.

2. Data Integration Process (ETL/ELT Flow)

- ETL Process:
 - Extract: Data is collected from multiple sources (POS, social media, website).
 - Transform: Data is cleaned and structured where necessary.
 - Load: Processed data is stored in the data warehouse for analytics.
- ELT Process:
 - Extract: Raw data (e.g., customer reviews, clickstreams) are collected
 - Load: Put into the data lake.
 - Transform: AI/ML models process (transform) the data when needed.
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- Flow Diagram

[POS Data] ---> [RDBMS] ---> [Data Warehouse] ---> [BI Dashboard]
[Customer Feedback] ---> [Data Lake] ---> [AI Sentiment Analysis]
[Web Analytics] ---> [Data Warehouse] ---> [Marketing Reports]
[Supply Chain] ---> [Data Lake + Warehouse] ---> [Forecasting System]